

PORTO VELHO, Brazil

WMO: 828240

Lat: 8.709S

Lon: 63.902W

Elev: 294

StdP: 14.54

Time Zone: -4.00 (W04)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	66.0	68.2	53.9	62.4	82.7	58.5	74.0	82.4	14.8	81.8	12.2	81.8	4.5	180	0.287

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	19.5	96.8	75.2	95.2	75.7	93.6	76.3	82.5	87.9	81.6	87.4	80.9	87.0	7.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	162.7	84.6	80.5	160.2	84.1	79.2	153.3	83.2	46.7	88.6	45.7	87.8	45.0	86.5	92.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.3	11.3	9.6	DB	61.4	99.1	3.5	1.8	58.9	100.4	56.9	101.4	54.9	102.4	52.4	103.7	
			WB	57.6	84.6	4.7	2.9	54.2	86.7	51.4	88.4	48.7	90.0	45.3	92.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	80.6	80.6	81.1	81.5	80.8	80.8	80.9	82.7	83.1	82.6	81.9	81.0
	DBStd	2.72	2.12	2.16	2.11	2.43	2.83	3.06	3.34	2.75	2.84	2.34	2.35	2.27
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDD50	11487	949	856	965	946	954	924	958	1013	994	1012	956	960
	CDD65	6013	484	436	500	496	489	475	493	548	544	547	506	495
	CDH74	53373	3697	3360	4016	4252	4196	4366	4882	5734	5476	5084	4414	3896
	CDH80	21414	1126	1004	1287	1457	1512	1915	2430	2971	2616	2182	1646	1268
Wind	WSAvg	3.2	3.3	3.2	3.1	2.9	3.1	3.0	3.2	3.2	3.4	3.4	3.2	3.1
Precipitation	PrecAvg	77.2	12.8	10.7	11.2	7.8	4.4	1.1	1.3	1.4	3.7	6.7	7.7	11.2
	PrecMax	124.4	23.7	17.0	20.5	14.1	9.1	5.4	5.6	3.9	8.1	12.2	12.4	23.3
	PrecMin	30.7	2.6	2.6	2.6	1.9	0.8	0.2	0.0	0.0	0.0	0.6	2.7	1.7
	PrecStd	25.0	4.8	3.9	4.3	3.4	2.2	1.1	1.6	1.2	2.3	2.4	2.6	5.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.8	92.1	93.0	93.5	93.4	94.9	96.9	98.9	98.5	97.0	95.0	93.3
		MCWB	79.1	79.2	79.7	79.7	78.1	76.2	73.5	72.7	76.7	77.6	78.8	79.5
	2%	DB	89.8	89.8	91.0	91.5	91.5	92.9	94.9	96.9	96.6	94.8	92.9	91.2
		MCWB	78.7	78.7	79.0	79.7	78.5	76.9	73.5	73.5	76.7	77.9	78.9	79.1
	5%	DB	88.0	88.1	89.2	89.7	89.7	91.3	93.1	95.1	94.8	92.9	91.1	89.3
		MCWB	78.7	78.5	78.9	79.6	78.6	76.5	73.7	74.0	76.8	78.1	78.7	78.7
	10%	DB	86.2	86.2	86.9	87.8	87.8	89.6	91.4	93.2	92.8	90.5	88.8	87.1
		MCWB	78.3	78.3	78.6	79.4	78.7	76.8	74.5	74.4	76.8	78.0	78.6	78.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.5	82.1	82.6	83.5	83.4	82.7	81.0	80.9	81.6	82.3	82.4	82.5
		MCDB	86.9	87.3	87.1	88.3	87.8	87.9	87.8	89.3	89.6	89.2	88.8	88.3
	2%	WB	81.0	80.8	81.0	81.8	81.3	80.7	79.3	79.4	80.4	80.9	81.1	81.0
		MCDB	85.9	86.1	86.4	87.1	86.5	87.3	86.9	88.5	89.0	88.1	87.6	86.9
	5%	WB	79.9	79.7	80.0	80.7	80.2	79.4	78.3	78.4	79.4	80.0	80.2	80.1
		MCDB	84.9	84.7	85.5	86.4	85.9	86.4	86.2	87.5	87.9	87.2	86.6	85.7
	10%	WB	78.9	78.9	79.1	79.7	79.2	78.5	77.2	77.4	78.5	79.0	79.2	79.1
		MCDB	83.7	83.4	84.3	85.2	84.9	85.6	85.2	86.2	86.9	86.3	85.3	84.4
Mean Daily Temperature Range	5% DB	MDBR	13.0	12.8	13.3	13.9	14.4	17.2	20.5	21.8	19.5	17.1	15.1	13.6
		MCDBR	15.2	15.2	15.9	16.1	16.7	19.7	23.0	24.4	22.2	19.9	17.8	16.3
		MCWBR	6.1	5.8	6.1	6.6	6.7	7.3	6.8	6.5	6.3	6.4	6.4	6.4
	5% WB	MCDBR	13.8	13.6	14.0	14.5	14.5	16.9	19.0	20.1	19.1	17.3	15.7	14.7
		MCWBR	6.6	6.2	6.6	7.1	7.4	8.3	8.3	6.9	6.7	6.8	6.8	6.7
Clear-Sky Solar Irradiance	taub	0.398	0.389	0.393	0.396	0.397	0.401	0.416	0.599	0.672	0.520	0.452	0.422	
	taud	2.507	2.541	2.538	2.497	2.483	2.471	2.402	1.861	1.712	2.124	2.324	2.425	
	Ebn at Noon	299	300	294	284	275	269	266	226	219	260	281	291	
	Edh at Noon	36	35	34	34	33	33	36	64	77	52	43	39	
All-Sky Solar Radiation	RadAvg	1417	1388	1429	1455	1415	1533	1661	1632	1574	1592	1486	1389	
	RadStd	73	83	68	73	87	99	110	128	138	61	75	91	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-4.21	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page